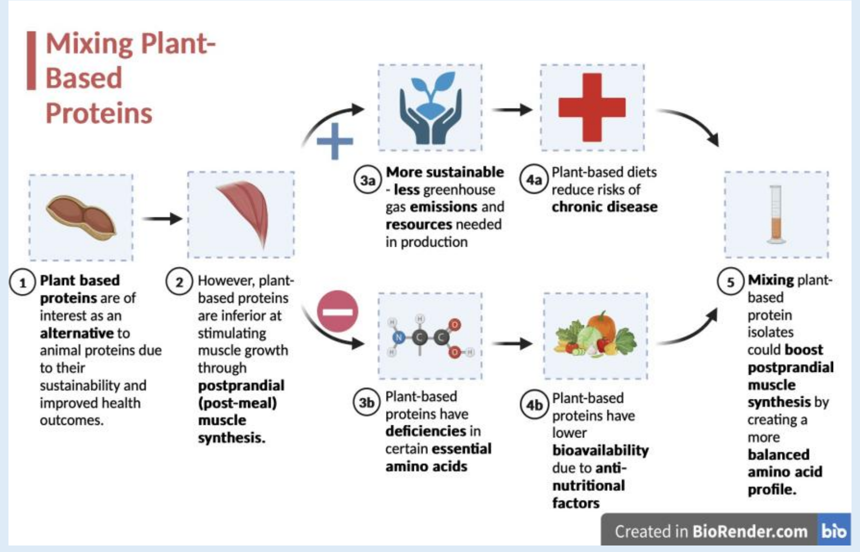
Using *Drosophila melanogaster* to Determine the Effects of Plant-Based Protein Isolates and Mixtures of Plant-Based Protein Isolates on Muscle Health, Development, and Growth

Background

Figure 1 – Background Visualization (from BioRender)



- **Plant-based proteins** are of interest as an alternative to animal- based proteins because they are **far more sustainable**; they emit less greenhouse gases, take up less resources and land in their production, and are more efficient energy-wise than animal- based proteins (Sabaté & Soret, 2014).
- However, they **are less effective at stimulating post-prandial muscle synthesis** as they often **have incomplete amino acid profiles** (Pinckaers et al., 2021).
- The body requires certain "essential" amino acids that it cannot synthesize by itself to be derived from the diet to effectively synthesize proteins and carry **out post-prandial muscle synthesis**, or the synthesis and repair of muscles following a meal.
- It has been proposed **that mixing plant-based protein isolates** can create a **more balanced amino acid profile** and allow them **to perform on par with animal proteins**; however, no study has examined postprandial muscle synthesis with **mixtures of specifically unenriched plant-based protein isolates** (Pinckaers et al., 2021).
- **The purpose of this study is to determine if mixing strictly plant-based proteins can help them perform on par with animal-based proteins in terms of postprandial muscle synthesis.**

Objectives

Hypothesis: The flies on a diet with mixed soy and potato protein isolates will have superior muscle function and health compared to those with pure soy or potato protein isolates because mixing plant-based protein isolates creates a more balanced amino acid profile that is conducive to post-prandial muscle synthesis.

Independent Variable: Concentration of soy and potato protein isolates in fly diets being tested.

Dependent Variable: Muscle function and health of the flies as a result of post-prandial muscle synthesis, measured by the flies' ability to climb.

Constants: Temperature and humidity at room temperature and 60-70% relative humidity; the gender and age of the flies, namely male and adult; and the strain of *Drosophila melanogaster* being used (wild-type).

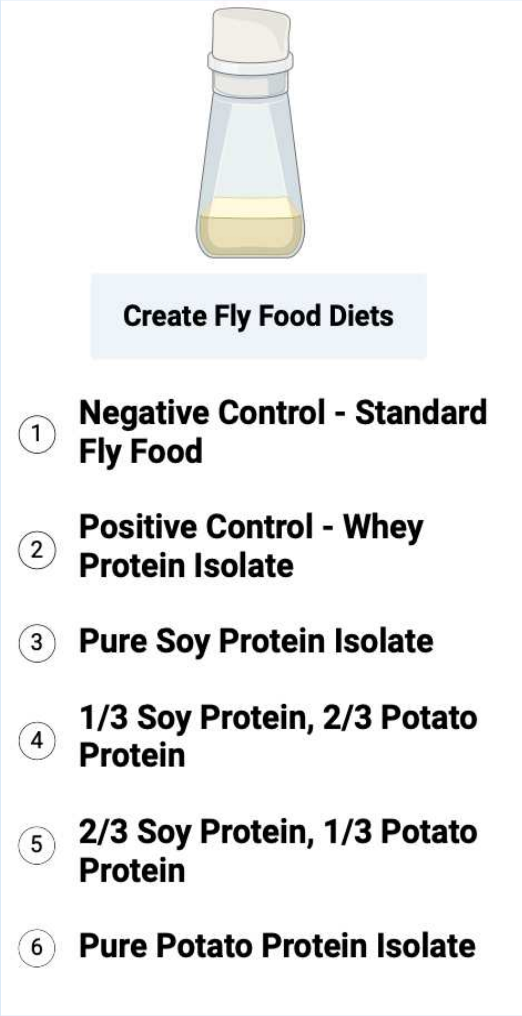
Control (-): Standard Yeast Diet

Control (+): Whey Protein Diet

Materials and Methods

Experimental Groups:

Figure 2 – Experimental Groups (from BioRender)



Climbing Assay:

1)Flies are tapped from their original vial into a climbing assay vial with an activity mark, or a line marking a height of 5 centimeters.
2)The flies are allowed to rest for one minute to allow them to adjust to their new environment.
3)A camera is set up to record the flies. The flies are tapped against a flat surface to force all of the flies to the bottom.
4)The flies will naturally climb back upwards due to negative geotaxis. It is recorded how many flies pass the activity mark in ten seconds.
5)10 trials with ideally 5-10 flies each for each experimental group were conducted.
6)Pass Rate is expressed as a percentage calculated by dividing the number of flies that passed the activity mark over the total number of flies in a given trial and multiplying by 100 to yield a percent value.

Figure 3 – Climbing Assay Pass Rate Formula

$$\text{Pass Rate} = \frac{\text{\# of Flies Past Activity Mark}}{\text{Total \# of Flies in Trial}} \times 100$$

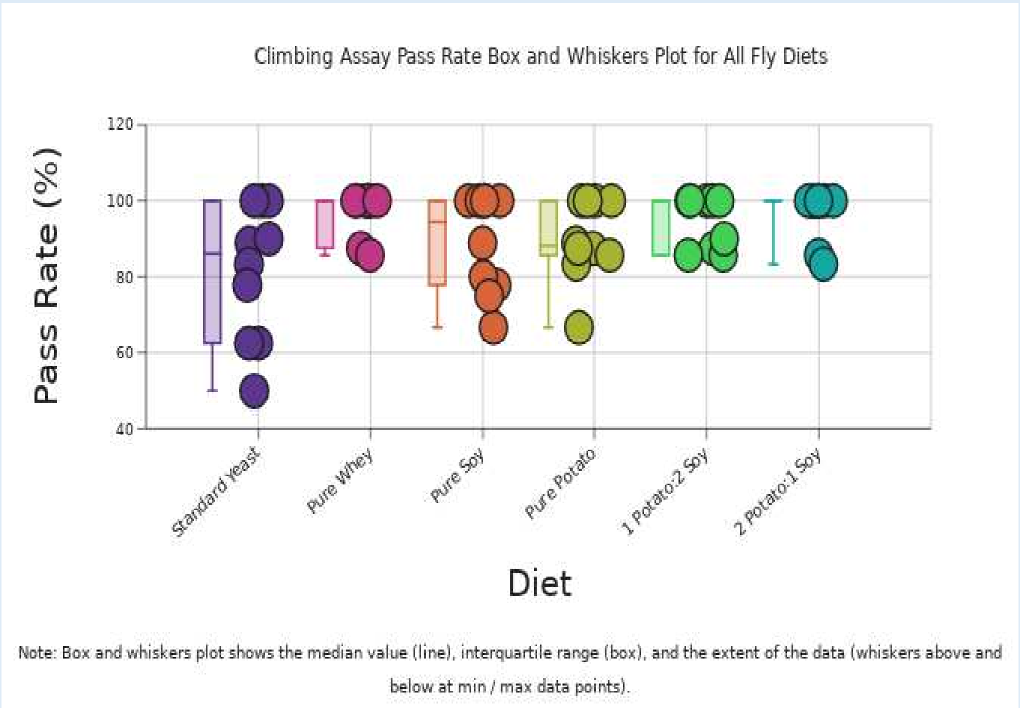
Results

Table 1 – Climbing Assay Data

Climbing Assay Processed Data (Pass Rate for All Experimental Groups)

Trial #	Pass Rate - Standard Yeast Diet	Pass Rate - Pure Whey Protein Diet	Pass Rate- Pure Soy Protein Diet	Pass Rate - Pure Potato Protein Diet	Pass Rate - 1 Potato:2 Soy Diet	Pass Rate - 2 Potato:1 Soy Diet
1	62.50	100.00	88.89	100	100	100
2	88.89	87.50	100.00	100	87.5	85.71
3	90.00	85.71	77.78	88.89	85.71	100
4	83.33	100.00	100.00	87.5	100	100
5	62.50	100.00	100.00	100	100	83.33
6	77.78	100.00	66.67	83.33	85.71	100
7	50.00	100.00	100.00	85.71	90	100
8	100.00	100.00	80.00	100	100	100
9	100.00	100.00	75.00	87.5	100	100
10	100.00	X	100.00	66.67	X	100

Figure 4 – Climbing Assay Box and Whiskers Plot (from DataClassroom)



Conclusions and Future Work

A Kruskal-Wallis statistical test was conducted on the climbing assay data. Since the p-value of **0.11** was greater than the significance of 0.05, there is **no statistically significant difference** in the climbing assay pass rates between groups. Thus, mixtures of plant-based protein isolates performed on par with animal-based proteins at stimulating postprandial muscle synthesis as climbing assay pass rates were statistically the same for mixture groups and the positive control of whey protein. However, what was unexpected was that there was also **no statistically significant difference in pass rates between the plant-based protein isolate groups of pure soy and pure potato protein and the positive control of whey protein** although it has been established that pure plant-based proteins are inferior to animal proteins at stimulating postprandial muscle synthesis. This indicates that there may have been a **high degree of error** as the results differed from those of previous studies. Therefore, the data is **inconclusive** and no conclusions can be drawn about whether mixtures of unfortified plant-based protein isolates can perform on par with animal based proteins.

The source of this error was likely a **decreased amount of data points**, which was because many flies died in the fly food prior to data collection in the climbing assay due to issues with the consistency and water retention of modified fly food diets. This reduced the number of flies, or data points, per trial and also reduced the number of trials in some cases where most of or all of the flies did not survive. Less data available to analyze increased the significance of outliers. Overall, the data was **inconclusive** and the validity of the hypothesis remains in question.

Future studies could reexamine this hypothesis to confirm the validity of the hypothesis using a larger quantity of flies and sample of data points to reduce the impact of outliers on the median data or by revising the method so that fly mortality rate is reduced in the modified fly diet vials.

References

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